RAG-based clinical QA dataset and model for Spanish

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Introduction

Correct extraction and interpretation of clinical information is essential for patient care, but manual extraction is time-consuming and prone to errors. Retrieval-augmented clinical QA models can be useful tools for automatized, updated and personalized consultation of medical guidelines. Both health workers and students can benefit from this kind of models, as long as they are reliable.

Goals:

1. Build a dataset of Spanish clinical guidebooks: SNS-1064
2. Develop a RAG-based QA model: SNS-RAG

All the code and the dataset are publicly available on GitHub:
`https://github.com/iker-gutierrez/SNS-QA-RAG`

Related Work

Retrieval Augmented Generation (RAG) improves factual grounding by extracting information from sources. **Asai et al. (2026)** demonstrate that:

- GPT-4o hallucinates scientific citations 78–90% of the times.

However, their RAG-based OpenScholar model achieves:

- a citation accuracy on par with human experts
- a reliable synthesis of scientific literature

In my model for Spanish clinical QA, I adopt their pipeline approach (but I do not adopt their evaluation method ScholarQABench, since it partially requires human evaluation, which goes beyond the scope of this project).

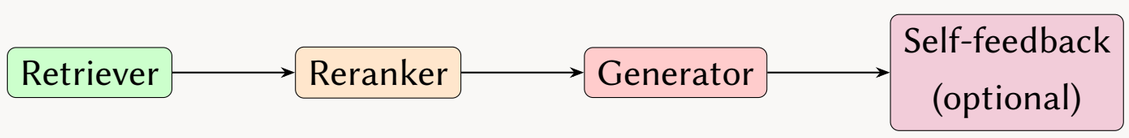
Dataset

The SNS-1064 dataset contains 6 Spanish clinical practice guidebooks from the SNS (Sistema Nacional de Salud / *Spanish National Health System*): anxiety.txt, diabetes.txt, ictus_management.txt, palliative_attention.txt, pediatric_palliative_care.txt, and secondary_prevention_of_ictus.txt. To turn these text files into a dataset, I captured the relevant information types using regular expressions and stored those information types into CSV columns:

#	**Columns**
1	Clinical topic
2	Clinical question
3	Subquestion (optional)
4	Judgement [short answer]
5	Scientific evidence
6	Additional considerations (optional)

- **Model's input**: columns 1-3 (topic, question, subquestion).
- **Model's output**: columns 4-6 (judgement, evidence, considerations).
- Total number of samples in the dataset: 1064.

Pipeline



Models:

- Retriever: multilingual-e5 + FAISS embeddings
- Reranker: multilingual MS MARCO Cross-Encoder
- Generator: Mistral-7B-Instruct (LLM)

Experimental Setting

- Train/test split: 80/20
- Retrieval: $top_k = 15$
- Reranking: $top_k = 3$
- Generation: prompting with instructions against hallucinations, e.g.

> You are a clinical assistant. Your task is to EXTRACT information from the context, not to invent or rephrase it. Rules:
> - Use ONLY information from the context
> - DO NOT add external information
> - COPY exact phrases whenever possible
> - Keep the language in Spanish
> - ALWAYS answer in this format:
>
> JUDGEMENT:
> EVIDENCE:
> CONSIDERATIONS:

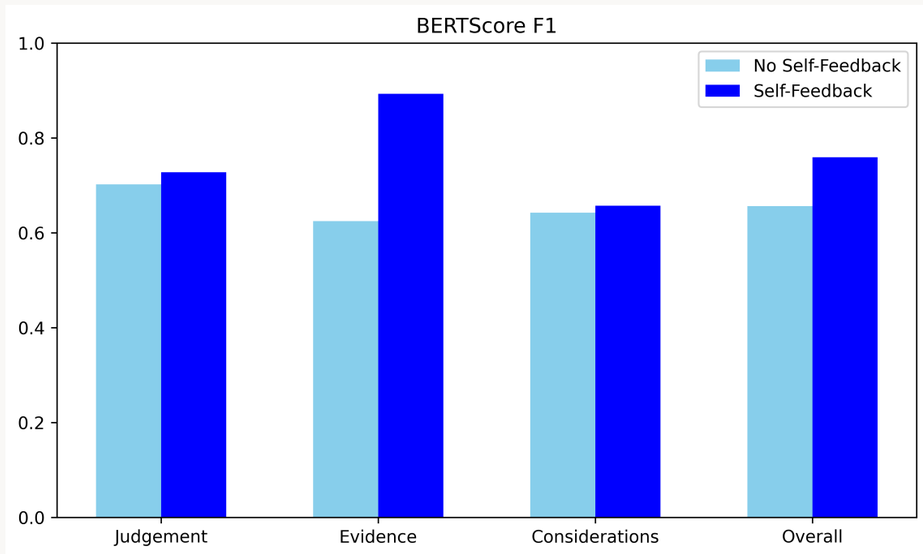
Evaluation Metrics:

- Cosine semantic similarity (mean).
- BERTScore (F1).

Results

- Self-feedback provides a major improvement in Evidence extraction.

Metric	Cos_sim	Cos_sim (+SF)	BERT	BERT (+SF)
Judgement	85.3	88.2	70.2	72.7
Evidence	81.8	95.0	62.5	89.3
Considerations	85.2	87.4	64.2	65.7
Overall	**84.1**	**90.2**	**65.6**	**75.9**



Conclusions

- The SNS-RAG model seems to provide a solid factual grounding for clinical QA.
- Instruction-based prompts and self-feedback seem to be essential to tackle hallucination.

Future work:

- Try other RAG-specific evaluation methods (e.g. RAGAS metrics).
- Try other reranking methods or other cross-encoder rerankers.

References

Asai, A., He, J., Shao, R. *et al.* (2026). Synthesizing scientific literature with retrieval-augmented language models. *Nature* 650, 857–863. `https://doi.org/10.1038/s41586-025-10072-4`